Sébastien Barbier

Staff Software Engineer | Full-Stack: TypeScript, React, Python

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GitHub | LinkedIn

Staff-level software engineer with 16+ years shipping production systems, including agentic AI platforms. Led frontend for an agentic AI platform at Unique AG and SAP FSM's move to microfrontends. Currently building Shellui (TypeScript, React, Python/Django) and improving my German beyond B1.

Personal Project

Full-Stack Developer Jan 2026 – Oct 2026

Shellui (shellui.com): open-source microfrontend shell *Zürich, Switzerland*

- Built a React/TypeScript microfrontend shell and CLI that gives apps one navigation and theme.
- Unified sign-in across apps with OAuth, SAML SSO, magic links and Supabase Auth.
- Designed multi-tenant PostgreSQL schemas in Django services, isolating each tenant's data.
- Built signed webhooks and Celery jobs with retries and an outbox, so failed deliveries are retried.
- Protected every release with 1,000+ tests and CI/CD on each pull request (web and Tauri).

Professional Experience

Staff Frontend Engineer Dec 2023 – Dec 2025

Unique AG *Zürich, Switzerland / Remote*

- Led frontend for an agentic AI chat platform used by financial services clients.
- Streamed LLM answers live in the UI over GraphQL, with structured outputs and tool calling in production.
- Integrated RAG and MCP into the chat flows, so answers use client context and stay compliant.
- Partnered with product, design and backend teams on delivery, keeping release quality high.
- Used AI coding agents in the daily workflow to ship frontend changes with strong test coverage.

Liquidity Provider, Decentralized Finance (DeFi) Aug 2021 – Nov 2023

Private investor (own capital) *Zürich, Switzerland*

- Provided liquidity to on-chain DeFi AMM pools, fully funding my living costs for 2+ years.
- In parallel, maintained personal open-source apps in React and Django REST Framework.
- Built FromEdwin.com as a learning project on asynchronous performance metrics, owning architecture, backend, observability and CI/CD.

Associate Development Architect Feb 2020 – Jul 2021

SAP (Schweiz) AG, FSM; promoted from Technical Lead *Zürich, Switzerland / Remote*

- As Technical Lead, moved FSM to microfrontends, aligning several teams on one architecture.
- Co-designed the FSM extensibility concept for secure third-party integrations, driving partner adoption with code samples and public webinars.
- Prepared management decisions on FSM's target architecture by assessing risks and effort.
- Designed, coded and tested complex FSM features, and helped others resolve complex issues.
- Mentored junior developers and coached the team on architecture, simplicity and best practices.

Web Technologies Consultant and Freelancer Sep 2016 – Jan 2020

Self-employed *Chiang Mai, Thailand / Remote*

- Built the UI of Mozilla's Notes by Firefox extension and contributed to its React Native app.
- Delivered chatbots, PWAs and web integrations for clients including Chefclub and Mila AG.
- Worked remotely and independently with international clients, from first call to release.

Front-end Developer / Technical Lead Jun 2014 – Aug 2016
Fluance AG *Solothurn, Switzerland*

- Led frontend in a four-person Scrum team building a cross-platform HL7 patient data interface.
- Owned documentation, testing and deployment; stack: AngularJS, jQuery and Foundation CSS.

Earlier Experience

Java Developer Sep 2012 – Jun 2014
Covalia *Besançon, France*

- Maintained Java HL7/DICOM exchange for a telemedicine platform under ISO 13485.

Part-time Lecturer Sep 2011 and Sep 2012
University of Franche-Comté *Besançon, France*

- Taught 98 hours of practical Java and HTML/CSS to third-year engineering students.

Co-founder and Full-stack Developer Jul 2009 – Jun 2012
Ionyse *Besançon, France*

- Co-founded a web agency during my master's, building and hosting client websites on a Python/Django CMS.

Technical Skills

AI
Agentic AI, Autonomous systems, LLMs, streaming responses, structured outputs, tool calling, RAG and MCP.
AI-assisted engineering with AI coding agents as part of the daily workflow. **Frontend**
TypeScript, JavaScript, React, Next.js, Angular, Tailwind CSS, PWA, accessibility, UX/UI.
Microfrontends, Redux, MUI, D3, streaming chat interfaces, React Native. **Backend & Architecture**
Python, Django REST Framework, Java, REST APIs, GraphQL, OpenAPI, SSO, Celery, PostgreSQL, Supabase.
Multi-tenant SaaS, microfrontends, event-driven services, signed webhooks, retries, Redis. **Delivery & Ops**
CI/CD, GitHub Actions, Docker, Cypress, unit and E2E testing, Sentry, Prometheus, Grafana.
Git, Linux server administration, Tauri desktop builds. **Domains**
Financial services, healthcare (HL7, DICOM, ISO 13485), enterprise SaaS, open source.
Field service management (SAP), telemedicine (Covalia), decentralized finance (DeFi).

Soft Skills

Ownership: built Shellui end to end, from architecture and UI to CI/CD and hosting.

Collaboration: aligned several SAP teams on microfrontends; partnered with product and design at Unique.

Mentoring: mentored junior developers at SAP and taught 98 hours of programming at university.

Communication: presented the FSM extensibility concept to SAP partners in public webinars.

Pragmatism & quality: simple, well-tested solutions shipped in small steps, backed by 1,000+ tests.

AI delivery: shipped agentic AI chat at Unique with streaming LLMs, RAG, and MCP for compliant financial-services clients.

Education

Master's in Computer Science 2011
Distributed Systems and Networks, University of Franche-Comté, France

Bachelor's in Computer Science 2009
University of Franche-Comté, France

Languages

French: Native | English: Fluent | German: B1 (Goethe-Zertifikat, July 2026)